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Can I design for play?

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Can I design for play? How pedagogues design for children's play situations in school

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ABSTRACT

The aim of this paper is to analyse how pedagogues are designing for children's play situations in schools. Based on design-based research and play design methods, the paper analyses three empirical examples using three theoretical concepts: play order, reflection in action and improvisation. A core argument of the paper is that the way in which pedagogues constantly observe, improvise and reflect on pedagogical practice is evident in how specific play designs are carried out. It is the capacity of professional pedagogues who understand play and children's perspectives and who consciously think about what they do while doing it, acting in the situation based on a specific reflection. This paper contributes to design for play activities, which are being developed in conjunction with the field of development within design practices and design research.

KEYWORDS

Play design, pedagogy, play order, improvisation, pedagogical reflection, school, after-school activities

Introduction

When professionals are designing for play in a school context, they often meet methodological and practical challenges, since schools historically tend to be focused on didactics of learning, which has led to a marginalisation of play in schools (Skovbjerg and Sand 2022). In the current paper, we state that designing for play can be a pedagogical approach that can meet the challenges facing professionals. In the Nordic countries, play has had an essential value within children's lives (Lillemyr 2003; Øksnes 2008; Ødegaard and Hedegaard 2020), and the pedagogical approaches in institutional contexts are often based on the possibilities for children to play and create play communities (Fleer et al. 2021); this follows the United Nations' Convention on the Rights of the Child (article 31) that specifies that every child has the right to play (UN 2013).

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Thus, understanding how pedagogues are designing for children's play in school is relevant for several reasons. In contexts such as schools, play is often seen as a didactic approach whose aim is to serve a particular learning purpose (Skovbjerg and Sand 2022), but in Denmark, since 2014, pedagogues have aimed to work in a school context with specific responsibilities in engaging children in play, participation and belonging to a community (Jensen 2016). In the present paper, we explore how pedagogues are designing for play by engaging children in the design process. Within the field of designing for play, inviting children to participate in the design process is well documented (Druin 2002; Page and Thorsteinsson 2017; Tonetto et al. 2020). Child-centred methodologies (Feder 2020) and participatory design processes (Simonsen and Robertson 2013) have been ways to reinforce children's participation, development and voice. The insights gained can be applied to educational and pedagogical practice to rethink and develop how pedagogues can work with play in contexts such as schools engaging and involving the children in these processes while broadening the variability of activities at school. Based on a design-based research project in two Danish schools among pedagogues and children 6–9 years old, the current paper seeks to answer the following research question: *How are pedagogies designing for play situations in schools? How are they engaging the children in the process of designing for play?*

We focus on how pedagogues handle design processes together with children in specific play situations in which there are specific designs. In doing so, we apply Lucy Kimbell's understanding of services as something enacted in relations and over time rather than 'a specific object to be designed' (Kimbell 2011, 42). Based on our understanding, this means that we need to distinguish between designing for play and play designing, which underlines the importance of the design processes. The current paper is not concerned with what play designs pedagogues plan for beforehand, but rather, the focus is on how they are involved in the design processes with children who are trying out what the pedagogues have planned for. A core argument is that the way in which pedagogues constantly observe, act and reflect on the pedagogical practice is evident based on how designing for play is carried out. In this sense, pedagogue's sensitive attunement and adjustments in designing for play are based on Schön and his idea of reflecting in action (Schön 1983), which details the capacity of professionals to consciously think about what they do while they are doing it, thereby acting in the situation based on a specific reflection.

> Much reflection in action hinges on the experience of surprise. When intuitive spontaneous performance yields nothing more than the results expected for it, then we tend not to think about it. However, when intuitive performance leads to

> surprises, pleasing and promising or unwanted, we may respond by reflecting in action. (Schön 1983, 56)

An aim of the present paper is to contribute to strengthening pedagogical practice through design methodology and processes, where improvisation is the core action when designing for play.

The current paper is structured as follows: First, we position our work in relation to the literature in the field of designing for play (see Appendix 1). Thereafter, we describe the *research context, methodology and design process* of the paper. The results are presented in the *analysis of play situations,* which is divided into three analytical sections: (1) *designing through play knowledge*; (2) *observing and waiting as a strategy when designing for play*; and (3) *designing through participatory sensitivity*. The analysis is followed by a discussion and conclusion.

Contextualising the pedagogue profession

The formal educational system in Denmark is a three-and-a-half-year bachelor's degree in social education, which gives the professional work title of pedagogue. The education is separate from Danish teacher education since pedagogues works within contexts such as youth clubs, day cares, preschools and institutions for people with special needs. An overall Nordic approach to pedagogy and children can be characterised as 'a child-centred, holistic approach that focuses on children's and parent's participation, autonomy, democracy and freedom where children's expressions, experiments and play are regarded as valuable' (Jensen 2016, 22). The pedagogues we focus on within the present paper are working in leisure time centres and supportive teaching activities at schools.

Literature review: The fields of designing for play

Designing for play is a broad interdisciplinary field. Different field of practices and research have contributed to the field for example; the field of architecture and environmental planning worked with designing for play for several years, focusing on playgrounds and urban environments (Hendricks 2011; Dattner 1974). As Hendricks concluded, to a greater extent, the approach to play design has been characterised as 'hands-on' guides for how to build environments for play (2011, 1). The current paper contributes to the field of designing for play through an attention to pedagogical practice (Skovbjerg and Sand 2022; Skovbjerg 2021, Jørgensen and Skovbjerg 2020; Skovbjerg et al. 2021; Bekker, Sturm, and Barakova 2010), but it is still vaguely thematised and developed, often emphasising that the importance of play has didactic learning potential (Skovbjerg and Sand 2022). As an approach to

design for learning throughout education, playful learning is an emerging field in which traditional instruction is being criticised. Within a knowledge society, we have embarked on a quest to teach new and broader skill sets to meet the challenges of the future (Jensen et al. 2021; Jørgensen 2022). Designing with the qualities of play can be a way to make this happen. The book *Framing Play Design* provides operational design guidelines for making specific play-based designs, as well as how to involve users and stakeholders in the process of play design. Jørgensen (2020) worked through narrative practices as being an approach to play design. Feder (2020) illustrated how play design can be driven through a child-centred design approach. In a paper published in the special issue *Learning, Breaking, Making: Analyzing Processes of Play* (Skovbjerg and Sand 2022), it was argued that working with play in school requires a specific theoretical understanding of the circumstances within which play is situated because schools, to a great extent, are embedded in the civilising logic of how to act, talk and play. Skovbjerg and Sand (2022) argued that the field of designing for play in school can add to a child-centred approach of pedagogical professionalism contexts but is vaguely developed theoretically and methodologically.

Within The Design Journal Tonetto et al. (2020) have a somewhat similar research interest when designing for the development of children's social skills through toy design. Their understanding of play was inspired by Vygotsky's developmental psychological perspective. We find it essential that design for play is understood as social practices to ensure that the support for play is closely related to the specific play situation created by children (Skovbjerg and Sand 2022). Furthermore, the present paper focuses on the design of pedagogical processes, not on products; it emphasises the importance of pedagogues being able to act situationally for the sake of children's play.

Design for and with children

In The Design Journal Can and Inalhan (2017) pointed out that children's environments are modelled on adults' values and needs, rather than on the needs of child users (see also Hart 1992; Rayner, Rayner, and Laidlaw 2012) and distinguished between designing *for* children and designing *with* children, where designing should be done to ensure the perspective of the child. This is very much in line with the epistemological aim of childhood sociology (James and Prout 1997) in which Scandinavian play research is rooted and from which the present research is grounded.

Theoretical framework: Play order, reflection and improvisation

The definitions of play can be difficult to formulate (Sutton-Smith 2001). In the current paper, we approach play through the concept of *play order*

(Skovbjerg and Sand 2022). Play order can be applied when aiming at the relatedness and connectedness that play has to the context in which it takes place. Based on Huizinga (1938), Bateson (1972) and Goffman (1986), *play order* (Skovbjerg and Sand 2022) means that play emerges and does not originate elsewhere but instead arises from the interactions and practices within the play order. Play is a way to relate to the world, other people, play stuff and oneself. As stated by Skovbjerg and Sand:

> … order is a logic on which actions (practice) are based, and which those involved (in this case) in a play activity have an idea about and which they seek to realise through their play actions without necessarily being able to define exactly what is going to happen. So it is through practice that the order emerges, while at the same time setting the direction for actions. (Skovbjerg and Sand 2022, 3)

When we are designing for play, the aim is for the play order to emerge and for the participants to be engaged in the play order (Skovbjerg and Sand 2022). The consequences of this understanding are, first, that the empirical work and analyses are concerned about following the play among the participants without judging the outcome in any way. Second, the playful activity is viewed as being driven by the participants, which means that the play order needs to be carefully nourished and continuously initiated by the play design. Finally, the role of the pedagogues involved in designing for play in specific play situations is concerned with supporting the play order and its continuous emergence, without specific outcomes and participatory-driven play. Overall, the pedagogue's most important action towards the play order is based on the emergence of what happens, 'in action', which cannot be predicted. In this sense, the design practice within the play order situation will lean towards improvisation, which relies on the ability to sense the situation and generate interplay between people, atmospheres, rhythms, bodies, roles, orders and contexts (Sand et al. 2021; Sand 2017). From this perspective, pedagogical improvisation when designing for play becomes an essential way to work with design.

Research context, methodology and design process

The paper is based on a research project entitled *Can I join in?—Play, Inclusion and Communities in School* (2018–2022) about designing for play in schools among children 6–9 years old (see Appendix 1). Two schools participated, with approximately 500 children and 40 pedagogues together with seven Danish researchers. Design-based research is a methodology (DBR) (Brown 1992; Barab and Squire 2004; Anderson and Shattuck 2012) in which design researchers and pedagogues, collaborate closely with design play activities. DBR is epistemologically rooted within pragmatism (Dewey 2008), aiming to make practical change (Barab and Squire 2004, 6). Designing for

play was structured through a design-based research model, inspired by Barab and Squire (2004); which underlies the importance of circularity and repeated iterations, including a sensitivity for local practical and theoretical context, generating new play design and reflecting upon them (Figure 1).

The process consisted of five iterations on five different play types, developed and scrutinised over a period of 10 weeks, as illustrated below (Figure 2).

***Context phase*:** At the beginning of the design project, an introduction workshop was held at the two schools. Pedagogues and researchers read

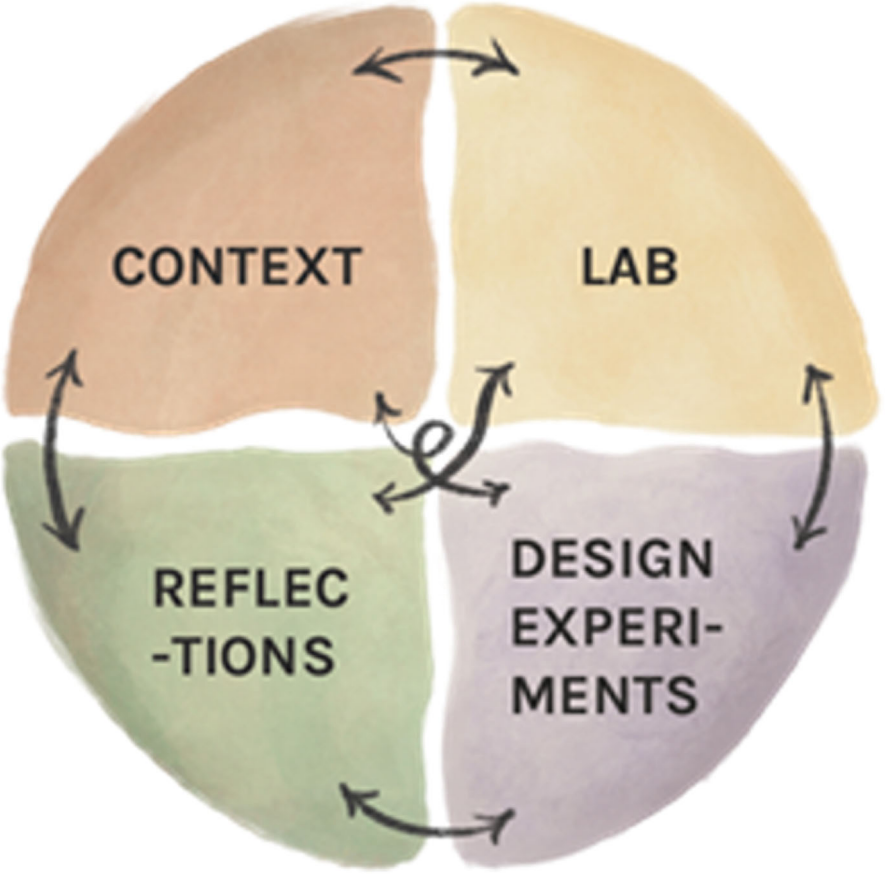


Figure 1. Design-based research model (illustrated by Emilie Bech Jespersen).



Figure 2. Operationalisation of the design-based research process (by Emilie Bech Jespersen).

and discussed texts on play. They tried out play activities to create embodied common experiences on different play types. Afterwards, each of the teams of pedagogues decided how they would organise their play experiment. In total, eight classes with 170 children (ages 6–9) and 15 pedagogues were chosen to participate. In the other school, the pedagogues decided to do the experiments in the Leisure Time Centre, where the children usually can choose freely between different activities that are planned by adults. Around 350 children attended the Leisure Time Centre together with 25 pedagogues.

LAB phase: For each design for play, the researchers initiated a LAB workshop in both schools. Two design tools were used to support ideation and planning. The first was an IdéKu, a tool for brainstorming developed by the network organisation Design to Innovate. The second was the Playful Spiderweb model (Figure 3), which was developed as a play design tool to support the didactical work and reflections. The pedagogues moved one or more ideas from the IdéKu into the Playful Spiderweb and made design decisions based on these ideas, moving them into more concrete design activities.

The Playful Spiderweb encourages pedagogues to reflect, have a dialogue about the design possibilities and be specific about organisational dimensions of play (Skovbjerg and Sand 2022; Jørgensen, Skovbjerg, and Eriksen 2021).

Play design experiment phase: The children and pedagogues joined and explored the play that the pedagogues had designed for and the pedagogues concurrently tried to make room for the children's play ideas and skills. The researchers conducted participant observation during the experiments.

Reflection phase: During each design for play, two reflection workshops were conducted and concrete experiences of pedagogues designing for play were discussed, often with input from the children. The pedagogues agreed on two questions that they frequently returned to throughout the project: How might we design for play pedagogically? How might we participate in children's play without taking control?

To generate data within the design phases, ethnography-inspired methods were used, including methods such as sensory and visual participation (Pink 2020), participant observation and fieldnotes (Hammersley and Atkinson 1983), sensory and emplaced interviews with children and pedagogues (Sand, Skovbjerg, and Tanggaard 2022), and audio recordings of play situations. Furthermore, the 32 reflection meetings were recorded, transcribed and analysed as empirical material. Concerning ethics, written consent from the participants was obtained from the children, parents and pedagogues in line with the Danish Code of Conduct for Research Integrity. The digital data processing programme Dedoose was used to analyse the empirical material.

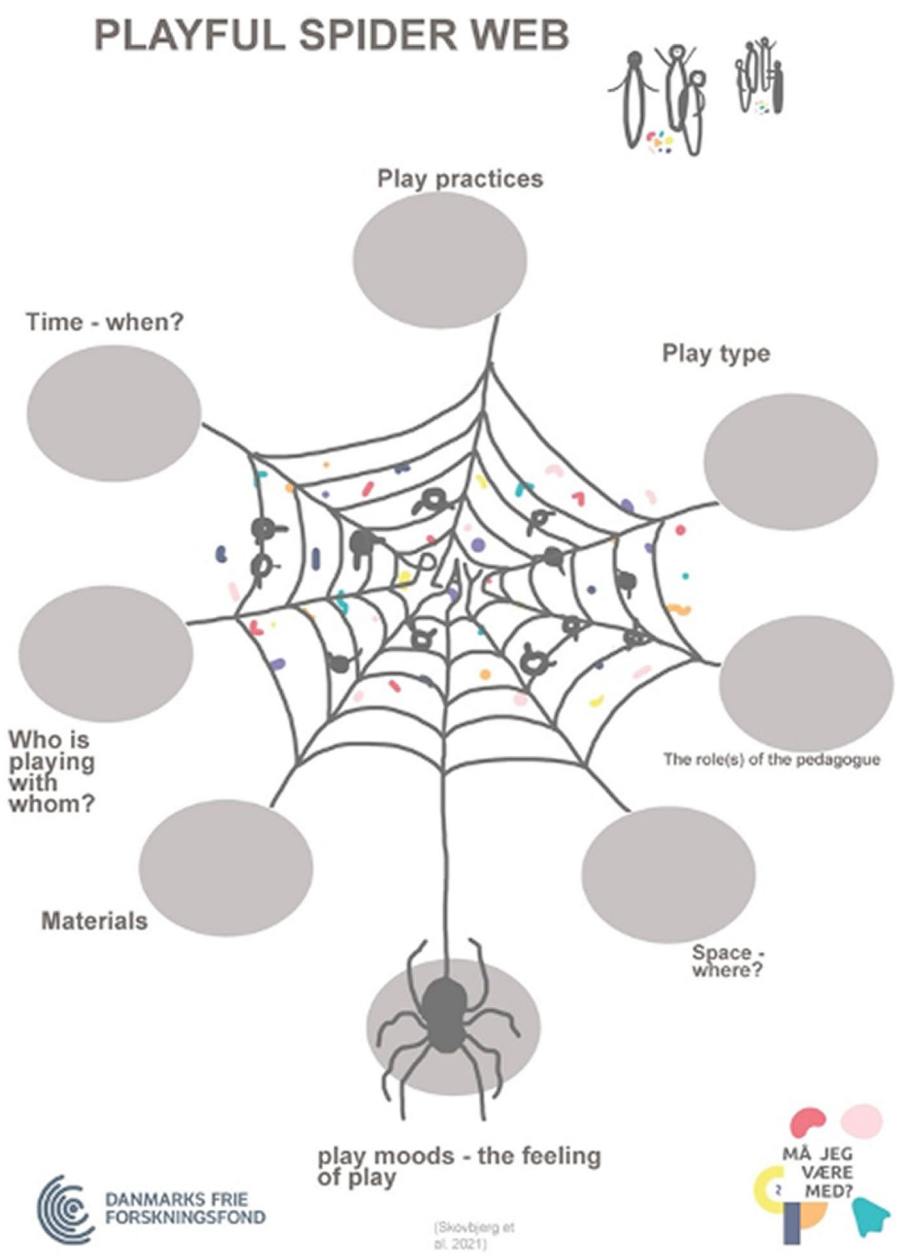


Figure 3. The Playful Spiderweb model. Pedagogues work directly with themes such as play practices, play type, the roles of pedagogues, space, time, play moods, materials and who is playing?.

The analysis strategy was inspired by grounded theory and consists of open and closed empirical readings to define codes and memos (Charmaz 2006), which are followed by 18 data-workshops within the research team. Because the two schools were influenced by COVID-19 regulations,

continuous dialogue occurred that dealt with ethics for example how the staff was reduced due to sickness and at the same time experimenting with play designs. This type of ethics is described as situational ethics, which specifies how researchers foster ethical reflections through careful observations (Hammersley and Atkinson 1983).

Insights based on the analysis of play

The 18 analytical workshops dealt with different types of coding, related to the four work packages of the project (Salmona, Lieber, and Kaczynski 2020). One workshop revealed how the pedagogue's way of supporting children's play was critical to how the children participated socially through play. This insight led to a specific data workshop related to how the pedagogues dealt with the design process during the play design experiments, which led to the following three analytical themes: *Play and moods: Designing with sensory attunement, Observing and waiting: A pedagogical strategy when designing for play and Designing with participatory sensitivity* (see Appendix 1).

Play and moods: Designing with sensory attunement

Four pedagogues designed the play design 'Pokémon Stratego'. They wanted to redesign a game that the children played inside and bring it outside and design it for a mood of intensity. In Pokémon Stratego, a red and blue team each get the same set of Pokémon cards, made by the pedagogues, and must compete against each other. The cards have different strengths and values, which afford different kinds of strategic thinking. A team leader for each team hands out a card to each person. The team has to run out, 'poke' a person from the other team and see who has the highest card. The team that has the most cards at the end wins. The pedagogical task is to make the children work together and collaborate strategically. The following are photographs taken during Pokémon Stratego, and below is an extract from the field notes when the children were running around while playing the game (Figures 4 and 5).

> The pedagogues Casper and Mads are preparing the red team for playing.
>
> Casper: 'Come here. Put your arms up and say, "Are we a team?" and "Yes, we are."'
>
> The girls are a bit unmotivated. 'You guys are a team. You pick a slogan. You get one minute'.
>
> They hesitate but step aside and choose 'Shahabkartata', which means sandal potato.
>
> Casper's voice is energetic, and he tries to keep their attention and get them eager to play. 'Now, you are going to go out and play at your maximum. Here are the rules (…)'.

Mads: 'The best card is Solgaleo, with 20 points. But Ditto can hit them all, but it dies, but that might be worth it if you get the other team's Solgaleo. So think strategically. You have to collaborate as a team. Try to figure out what cards the others have and communicate that to your teammates.'

Casper: 'Are we a team?'

'Yeah', they say tiredly.



Figure 4. Photographs taken during the play design 'Pokémon Stratego'. Photo: Sand.



Figure 5. Photographs taken during the play design 'Pokémon Stratego'. Photo: Sand.

Casper: 'Then raise your arm'.

Together, 'Shahadkaratata', they say. 'Sandal-partatoe', they say in Danish and laugh.

The game begins. Something happens to the tired girls. At once, an intensity arises. The children change their way of moving, looking and talking; their eyes zoom out, and they focus and scan the bushes for opponents. They move differently, too, as

> they bend their knees, run on tiptoes and stand and run closer together. After 45 minutes, Casper yells, 'Here comes Solgaleo!' Four to five children run forward. The mood of the situation becomes almost chaotic, which is sensed through the way the children hesitate, tiptoe back and forth, observe and manoeuvre in relation to the others' moves. A girl, Iben, runs beside the researcher, and the researcher asks Iben, 'What card do you have?'
>
> She identifies her card to me and says, 'Shhhh'. She runs and stops suddenly as she sees the other team enter the field. She runs backward. Two of her teammates join her; they identify their cards and secretly place them in their pockets. Later, Iben gets into a duel. Iben shows her card, and that sparks an echo of sound on the battlefield: 'She has Solgaleo'. Iben runs back to the team leader with the card she just won.
>
> A boy from the other team shouts, 'I have to kill Iben'. The others want to defend her, run forth and get into a battle with the others.
>
> A boy from the other team is wild. 'After Iben!'
>
> A boy from the blue team says, 'What the hell are you doing, Ali?' The mood is chaotic. The children are wild. One team tries to catch Iben and get Solgaleo, while the other team tries to protect Iben.
>
> Pedagogue Casper yells, 'Remember to talk together. Remember to work together; a strong and vague card can work together' (Fieldnotes 03.12.2020).

The example shows how the pedagogues act within a specific play design situation, meeting the participants and play order with which they are about to get involved in an improvisational way. Based on the context phase, the pedagogues and researchers read pedagogical theory together. One paper was about moods in play, which inspired the pedagogues to design for moods together with children. Casper observes how the girls seem unmotivated to play; he tries to motivate them by creating a name for their team and by engaging in the design process using a high tone and energetic voice to reinforce the intense play mode. As in improvisation, one manoeuvres within a regulated pattern and acts upon what happens, making different sensory dimensions complement each other. The pedagogues could not have predicted the play order in advance, so the improvisational sensitivity towards what is happening between the children allows the pedagogues to maintain the play order, that is, the logic of play between the children. This becomes important when they let loose the Solgaleo card, which can take out all the other cards. Here, the improvisational pedagogical task balances between maintaining the play order and ensuring that the intense, at times chaotic, mode does not make some children angry or sad. For example, this is done by telling them to be gentle to each other, telling them to talk and work together and putting an arm around a child to clarify rules.

Observing and waiting: A pedagogical strategy when designing for play

In the following empirical material, the pedagogue has designed the play design 'Stone workshop', which had open structures and was led to the children to define. The play design takes place outside the school in an area in an open wooden shed with benches and tables. For three weeks, about 8–12 children participated, some briefly because their parents picked them up, some for two hours. Figures 6 illustrate the children playing.

> 'Do you want to play with stones?' The pedagogue, Jess, asks children who are close to the play area or some who walk by. Five children say yes, and they join in.
>
> 'What are we supposed to do?' another child asks.
>
> The pedagogue does not explain anything but starts collecting stones and more stones, putting them at the table. The children start collecting stones, too. They collect different sizes and the number of stones on the table increases. Jess, alongside the table, picks up a stone, touches it and moves it from hand to hand.
>
> 'This one looks like an old arrowhead'.
>
> 'JESS, JESS WE FOUND A REALLY BIG STONE', Per and Rune shout.
>
> Jess walks with them to a huge stone, and he starts pretending that he will move it. The boys start helping him. 'Ah, it is big', Per says. 'Maybe too big to bring to the table', Rune continues. They touch the stone again and again, push it and laugh. They move back to the table with Jess. Luka, another boy, hits two stones



Figure 6. Picture from the play design 'Stone workshop'. Photo: Jørgensen.

> against each other again and again, but it is difficult to do because the stones have several surfaces, and it is difficult to find the right places to hit and break them.
>
> Kasper helps him find another stone to destroy, touching the surfaces and showing him how to look for good ones. 'Maybe this one', Luka says, showing the stone to Jess. 'Yes, give it a try'.
>
> Luka starts hitting the stone and it breaks. 'Yes!' He is happy with the breaking (Fieldnotes 19.04.2019).

In this example, the pedagogue invites the children to participate by asking if they want to join. When they ask what they are supposed to do, he starts collecting stones as a way to create a pathway to play design, even though he does not explain anything. He is practicing the play order (Skovbjerg and Sand 2022), showing them how they can be part of the confirmation of meaning in a specific situation. He observes what happens when he starts collecting and shows them what it is all about; he is making his next decisions based on this observation and waiting for the children's feedback on his design. This illustrates an interactive design process between pedagogues and children. When they start collecting stones, as he does, it signifies that they understand his play order, and he touches and talks about the stones, showing the children how the play order is practiced. While collecting, they find a really big stone. He follows the children to the stone, and instead of saying that there is no chance that they will be able to carry the stone to the table, he maintains the intention of the play design and gives it a try together with the boys. He touches the stone, and when he is pretending, he imagines that it is possible based on the boys' idea and the intentions of the play design. The pedagogue balances between constituting the play design through actions and observing and waiting to explore possibilities coming from the play order initiated by the children participating.

Designing with participatory sensitivity

The following example is a play design based on Dungeons & Dragons (D&D), where the pedagogue Thomas is a gamemaster (Figure 7).

Jacob wants to figure out who kidnapped the mayor. He thinks it was the wizard. After a long dialogue with the gamemaster, he throws the dice to determine whether the wizard is telling the truth. Unfortunately, Jacob makes a bad throw with the dice, which only indicates that the wizard tells the truth, but there is still a risk that he is lying. Jacob tries thoroughly to make the wizard admit that he kidnapped the mayor, but the wizard refuses. Later, the following is observed:

> Three other children have fought against some snowmen for a while, in the end conquering the snowmen. Now, Jacob is impatient because he is standing in another part of the imaginary world where nothing is happening; he negotiates



Figure 7. Pictures of the children and gamemaster playing D&D. Photo: Jensen.

with the gamemaster and buys a reindeer. With a big, challenging smile, he says he wants to ride fast to the others but that he does not want Charlie (a younger boy) to ride with him. Charlie smiles and seems thrilled about the challenge, and in return, he threatens Jacob with his 'burning hands'. Jacob replies to Charlie's threat by drawing his bow. 'I'll shoot you'. The fight is on! They will let the dice decide who won the battle.

The other players are observing the fight and laughing and making comments. Charlie attacks Jacob. The tension increases. The children are excited; they laugh and jump around the floor. Charlie throws the dice and hits Jacob. Now, he has to work out how much he damages him. Because of the magic force in the 'burning hands', he can throw the dice four times. First, he hits six, which results in huge applause from Jacob. Hereafter, he hits four and then two times six. The children laugh even more. The game has gone completely crazy. Jacob starts by having 13 lives but loses 20 lives because of Charlie's attack, which positions him at -7 lives. That means that Jacob is unconscious (when you have -10, you are dead). The sound increases. Until this point, everyone has found the play situation fun, but Jacob now becomes pitiful because he finds out how unusually hard he has been damaged. The gamemaster, Thomas, interferes and says that Charlie and Jacob do not have time for more today. The other children enter a cave where they find the mayor (who has a long white beard). They rejoice because they have found him and ask the gamemaster, 'Who kidnapped him?'

'It was the wizard', the gamemaster says.

Now Jacob yells, 'Yes, I knew it. That was what I was trying to get the wizard to admit'. Everyone laughs.

The gamemaster: 'We will finish for today. Now, please help me clean up'.

'I knew it, I knew it', Jacob says (Fieldnotes, 27.11.2019).

The pedagogue encourages the children to cooperate, but in the above situation, the dynamics of the play situation change radically. Suddenly, the children are not cooperating about solving a common quest but fighting against each other. The children have almost nothing to win and everything to lose—they are deeply involved in playing. The fate of the dice roll forces the pedagogue to improvise if Jacob does not feel too bad about losing face with the younger boy, Charlie. By letting Jacob be right in his assumption about who kidnapped the mayor, the pedagogue is giving Jacob a symbolic victory and returning him to a central role in the play situation, making sure that he still finds the play order meaningful. The example illustrates how important it is to be sensory attuned in design processes because small, often unnoticeable, signs of communication or aspects within social complexities are valuable to the participants. The pedagogue manages to improvise and, at the same time, respect the meaning of the play order while creating a new worthy position in which Jacob can participate. From a narrative perspective, he also manages to tie an elegant bow to the whole play design by returning to an earlier challenge in the play activity. This emphasises how the design process can be planned to some extent but also is an iterative process where exchange and meaning happens *in situ* and that designers as pedagogues need to be able to improvise within that space.

Discussion

The initial aim of the current paper was to analyse the following: *How do pedagogues design for play situations in schools? How are pedagogues engaging children in the processes of designing for play?* We have addressed these questions through descriptive empirical examples of play situations. The three examples' show how designing for play fosters situations that are characterised by uncertainty and unpredictability. Recent studies have nuanced how play in schools in a Nordic context is often marginalised or related to a learning purpose instead of seeing play as having a value in and of itself (Øksnes et al. 2020). With the current paper, we argue that designing for play reinforces that children's expressions, experiments and play are regarded as valuable. Observations, interviews and reflection meetings (Figure 2) have generated knowledge about how the pedagogues in schools can have difficulties interfering with and framing children's play because of a pedagogical tradition that values free play. Furthermore, the design-based research process has proved to be useful because it systematically involves

pedagogues in a design phase and guides them to design for play. Brainstorming by using the IdéKu and coming up with their own play designs seemed to be a way to break free from the idea of going 'behind' children's play and instead going 'in front' of the process (Jørgensen 2022). The play design experiments were influenced by more practical everyday circumstances at the schools, as sickness among the pedagogical staff and meetings and as researchers we had to acknowledge that practical circumstances influenced the complexity, dept, repetition of play designs. Because the project spanned over one and a half years, the pedagogues worked through five DBR processes, which gave them experience in using the tools IdéKu and Spiderweb and could anticipate the methodological process when designing for play.

An essential factor in this design-based research project has been that pedagogues have been given the time to prepare their play design, which has given them some kind of predictable structure. However, at the same time, play design has an embedded uncertainty. Akama, Pink, and Sumartojo (2018, 35) described an element of design processes as *learning through doing*. They referred to Ingold and emphasised that design processes are 'an art of enquiry' being in the open and being in a process of experimentation that follows what unfolds. Improvisation is a practice of relying on surrounding structures, exploring potentials, retrospective sensemaking and seeing mistakes as a source of learning (Sand et al. 2021). Observing how the pedagogues in the three empirical examples acted in play design situations, it becomes clear that they supported the children's play in a specific way: observing, waiting, reflecting and operating with sensitivity towards the specific child and participatory opportunities the play design afforded. Being observant and sensitively tuned into the play and the children were necessary for, for example, maintaining a certain play mood and supporting the play participation of the children, as demonstrated in the empirical examples.

The qualitative data have shown that design for play can reinforce and develop new approaches for pedagogues designing for play in schools and contributing to an ongoing critical discussion and analysis of how design-based research can develop or rethink different fields of pedagogical practices. Several studies have focused on designing for play as a situated practice between two or more peers (Stivers and Sidnell 2016), whereas only a few studies focus on designing for play among children and adults. Pursi and Lipponen emphasised that educational studies need research designs that understand how children and adults interact in play to bring forth more multidimensional practices of adult–child–group interactions (2018, 22). They argued:

'Current play research needs expanded methodological tools and more nuanced and context specific analytical terminology to first, detect joint play from the flow of interaction more systematically; and second, to observe and analyse the organisation of play interaction as collective multiparty activity between adults and children' (Pursi and Lipponen 2018, 23).

We argue that design for play can help 'expand' the methodological tools that Pursi and Lipponen strived for.

Conclusion

The current paper has analysed how pedagogues can design for children's play situations in schools. Based on design-based research and play design methods, the present paper analysed three empirical examples. The three theoretical concepts used were play order (Skovbjerg and Sand 2022), reflection in action (Schön 1983) and improvisation (Sand et al. 2021). A core argument of the paper is that the way in which pedagogues constantly observe, act and reflect on pedagogical practice is evident in how specific play designs are carried out. It is the capacity of professional pedagogues who understand play and children's perspectives and who consciously think about what they do while doing it, acting in the situation based on a specific reflection. The current paper is contextualised within a Nordic pedagogical tradition, where play is seen as a highly important practice with value in itself. Designing for play related to a Nordic pedagogical tradition is an interdisciplinary field in development, and we encourage scholars internationally to engage in future research that develops design-methodological approaches and epistemological understandings.

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Appendix 1. List of all play types and play designs.

Play type	Different play designs	Pedagogues involved in specific play designs
Creative play	Quicksand	3
	Miniature landscape	2
	The open cabinet	2
	Nature colours	1
	Play with clay	5
	Word relay	2
	Telling from figures	2
	Boat building	1
	Stone workshop	1
Role play	Cafe play	2
	Medieval play	2
	Circus play	3
	Dramatic play	1
	Family play	3
	Mini-city	2
	Dungeons & Dragons	1
	Teddy bear club	3
	Figure construction	2
Construction play	Trolls	6
	Insect hotel	2
	Movie construction play	3
	Build a city	3
	Fantasy landscape	2
Dramatic Play	Theatre through dices	1
	Play a story	2
Movement Play	Pokémon Stratego	3
	Nerf gun war	2
	Olympic Games challenges	3
	Ninja warriors	1
	Vying in school	4
	Vying outside school	3